InternXpert Smart Internship Explorer and Verification Portal

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Abstract: The rapid expansion of online internship platforms has created challenges related to data authenticity, security, and efficient coordination between students and organizations. To address these issues, this paper presents Intern-Xpert – a Smart Internship Explorer and Verification Portal, an intelligent web-based system designed to streamline internship discovery, application management, and verification processes. The proposed platform integrates the Gemini AI API to assist users with intelligent internship search, resume analysis, and automated communication. Intern-Xpert consists of three core modules: Admin, Student, and Company. Administrators monitor system activities and verify the authenticity of registered students and companies. Students can register, manage profiles, upload resumes, search and apply for internships, track application status, receive email notifications, and attend interviews. Companies can post internship opportunities, review applications and resumes, schedule interviews, communicate with candidates, and update selection status. To ensure data integrity and privacy, Blockchain technology is utilized to store sensitive information securely, providing immutability and resistance to tampering. The system is implemented using React.js for the front-end, Node.js with Express.js for the back-end, and MongoDB for database management. By combining AI-driven assistance with blockchain-based security, Intern-Xpert enhances transparency, reduces fraudulent activities, and improves overall internship recruitment efficiency.

1 INTRODUCTION

The increasing demand for practical work experience has led to a significant rise in online internship opportunities. While digital platforms have simplified access, they also present challenges related to data authenticity, security, and efficient coordination between students and organizations. Conventional internship management often involves manual verification, inconsistent communication, and difficulty tracking application status, which can result in delays, mismanagement, and fraudulent activities. Addressing these issues requires a system that ensures both intelligence in internship management and robust data security.

To meet these requirements, this paper proposes Intern-Xpert, a Smart Internship Explorer and Verification Portal. Intern-Xpert is a web-based platform designed to streamline internship discovery, application management, and

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verification processes while ensuring data integrity and transparency. The system leverages the Gemini AI API to provide intelligent internship search, resume evaluation, and automated communication, enhancing user experience and reducing manual effort. The platform is divided into three primary modules: Admin, Student, and Company. Administrators monitor system activities and verify the authenticity of registered students and companies. Students can create profiles, upload resumes, search and apply for internships, track applications, receive notifications, and attend interviews. Companies can post opportunities, review applications, schedule interviews, communicate with candidates, and update selection status.

To ensure secure and tamper-proof storage, Blockchain technology is used specifically to store sensitive information such as student profiles and interview schedules. This approach guarantees immutability, prevents unauthorized modifications, and improves trust between stakeholders. The system is implemented using React.js for the front-end, Node.js with Express.js for the back-end, and MongoDB for database management, providing scalability and performance. By combining AI-driven assistance with blockchain-based security, Intern-Xpert enhances transparency, reduces fraudulent activities, and optimizes the overall internship recruitment process.

2 LITERATURE REVIEW

The increasing adoption of online internship and recruitment platforms has encouraged the development of digital systems to manage applications and communication efficiently. Early internship management systems focused mainly on automating registration and application processes. In, a centralized web-based internship portal was proposed to simplify student–company interactions, but it lacked mechanisms for secure data verification and fraud prevention.

Artificial Intelligence (AI) has been widely applied in recruitment systems to improve candidate matching and decision-making. Research in introduced an AI-based job recommendation system that analyzes user skills and preferences to suggest relevant opportunities. Similarly, resume screening using machine learning techniques was discussed in, where automated analysis reduced recruiter workload. However, these systems did not address data integrity and trust in user-provided information.

Security and authenticity issues in digital platforms have led to increased interest in Blockchain technology. In, blockchain was used for secure storage of academic credentials, ensuring immutability and transparency. A blockchain-based recruitment framework was presented in, where candidate data was protected against unauthorized modifications. Despite improved security, these systems lacked intelligent assistance and user-friendly application management.

Hybrid approaches combining AI and blockchain have also been explored. In, a decentralized hiring platform utilized blockchain for data security and AI for candidate evaluation. Another study in emphasized the role of smart contracts in automating recruitment workflows. However, these systems did not specifically support internship-focused functionalities such as interview scheduling and application tracking.

Web-based platforms using modern technologies were discussed in and, highlighting the effectiveness of React.js and Node.js in scalable recruitment applications. Additionally, email notification and real-time communication mechanisms were examined in to improve coordination between applicants and recruiters. Existing literature reveals a gap in integrating AI-driven assistance with blockchain-based verification for internship management, which the proposed InternXpert system aims to address.

Several studies have also examined the limitations of centralized recruitment platforms with respect to scalability and single points of failure. In, the authors discussed how centralized databases increase vulnerability to data breaches and unauthorized access, particularly when handling sensitive user information such as personal profiles and interview schedules. This limitation highlights the necessity for decentralized storage mechanisms to enhance system reliability and security.

User experience and transparency have been identified as critical factors in the adoption of digital recruitment systems. Research in emphasized the importance of real-time application tracking and status notifications to improve candidate engagement. However, many existing platforms rely on manual updates, which can lead to delayed communication and reduced trust between applicants and recruiters.

Verification of user authenticity remains a major challenge in online internship systems. In, a digital identity verification model was proposed to validate user credentials, yet the approach required third-party intervention, increasing operational complexity. Blockchain-based identity management

solutions, as discussed in, offer decentralized verification but often lack integration with intelligent recommendation systems.

Recent advancements in AI-driven conversational agents have improved automated communication in recruitment platforms. The study in demonstrated the effectiveness of AI chatbots in handling applicant queries and scheduling interviews. Despite improved responsiveness, these systems typically operate independently of secure data storage frameworks, raising concerns about data consistency and integrity.

From the extended review, it is evident that existing solutions address specific aspects of internship and recruitment management, such as automation, recommendation, or security, in isolation. A comprehensive system that integrates AI-powered internship exploration, automated communication, and blockchain-based storage of sensitive data including profiles and interview schedules remains largely unexplored. The proposed Intern-Xpert platform aims to bridge this research gap by providing a secure, intelligent, and transparent internship management solution.

3 EXISTING SYSTEM

At present, internship management is carried out through a combination of disconnected online platforms, university portals, and email-based communication channels. This fragmented ecosystem leads to considerable inefficiencies for both students and recruiting organizations. Students are required to navigate multiple websites and communication sources to discover opportunities, submit applications, and track responses. This scattered process often results in confusion, missed deadlines, and a lack of clarity regarding application progress. Recruiters, on the other hand, must handle large volumes of applications manually, making the process time-consuming and difficult to manage efficiently. Another major limitation of the existing system is the absence of centralized data management and intelligent matching capabilities. Most current platforms function only as listing services rather than comprehensive recruitment solutions. They fail to leverage student skill profiles, academic performance, or career interests to recommend relevant internship opportunities. As a result, students frequently apply to positions that do not align with their qualifications, while companies receive numerous unsuitable applications. This mismatch reduces recruitment efficiency and increases the workload for hiring teams, ultimately slowing down the selection process. Furthermore, existing internship management systems lack robust security and verification mechanisms. The absence of integrated credential validation makes it easier for fraudulent postings and false applicant information to circulate within the system. Sensitive personal and organizational data is often stored in inadequately protected databases, exposing it to potential data breaches and misuse. These security and privacy concerns diminish trust among users and discourage active participation from both students and employers. Consequently, the current approach not only suffers from operational inefficiencies but also faces serious challenges related to data integrity, authenticity, and user confidence.

4 PROPOSED SYSTEM

The proposed system, Intern-Xpert, is a Smart Internship Explorer and Verification Portal designed to overcome the limitations of existing internship management platforms by integrating Artificial Intelligence and Blockchain technology. The system aims to provide a secure, transparent, and efficient environment for students, companies, and administrators involved in the internship recruitment process.

Intern-Xpert follows a modular architecture consisting of three primary entities: Admin, Student, and Company. The Admin module is responsible for monitoring overall system activities, managing user accounts, and verifying the authenticity of registered students and companies. This verification process helps eliminate fraudulent profiles and ensures platform credibility. The Student module allows users to create and manage profiles, upload resumes, search for internships using intelligent recommendations, apply for opportunities, track application status, receive email notifications, and participate in scheduled interviews. The Company module enables organizations to post internship opportunities, review student applications and resumes, schedule interviews, communicate with candidates, and update selection results.

To enhance intelligence and automation, the system integrates the Gemini AI API. AI-driven functionalities include smart internship recommendations based on student skills and preferences, resume analysis to assess candidate suitability, and automated communication to streamline interactions between students and

recruiters. This reduces manual effort and improves decision-making efficiency.

For secure and tamper-proof data management, Blockchain technology is employed to store sensitive information such as student profiles and interview scheduling details. By leveraging blockchain's immutability and decentralization, the system ensures data integrity, prevents unauthorized modifications, and increases trust among stakeholders.

The system is implemented using React.js for the front-end to provide an interactive user interface, Node.js with Express.js for the back-end to handle business logic, and MongoDB for scalable database management. Through the integration of AI-driven intelligence and blockchain-based security, Intern-Xpert offers a reliable and efficient solution for modern internship recruitment.

5 METHODLOGY

The methodology of the proposed Intern-Xpert system focuses on integrating intelligent automation with secure data management to ensure an efficient and trustworthy internship recruitment process. The system follows a structured workflow that enables seamless interaction between students, companies, and administrators. Initially, users register on the platform based on their respective roles: Student, Company, or Admin. During the registration phase, essential details are collected and validated. The Admin module performs verification of registered students and companies to ensure authenticity. Once verified, user profiles are activated. Sensitive profile information is securely stored using Blockchain technology, ensuring immutability and preventing unauthorized modifications, while non-sensitive application data is maintained in the MongoDB database. After successful registration, students create and manage their profiles by uploading resumes and specifying skills and preferences. The Gemini AI API is utilized to analyze resumes and generate intelligent internship recommendations based on skill matching and relevance. Students can browse recommended internships, apply for suitable opportunities, and monitor application status through the web interface. Companies post internship opportunities by specifying required skills, duration, and eligibility criteria. Submitted applications are reviewed through the company dashboard, where resumes are evaluated with AI assistance to improve candidate shortlisting. Companies can schedule interviews for selected candidates, and interview scheduling details are securely recorded on the blockchain to maintain data integrity and transparency.

The system supports automated communication through email notifications and AI-assisted messaging. Notifications are triggered for key events such as application submission, interview scheduling, and selection updates, ensuring timely communication between stakeholders. Throughout the process, the Admin continuously monitors system activities, manages user access, and resolves disputes if necessary. By combining AI-driven intelligence with blockchain-based security and a scalable web architecture using React.js, Node.js, Express.js, and MongoDB, the proposed methodology ensures transparency, reduces fraudulent activities, and enhances the overall efficiency of internship management.

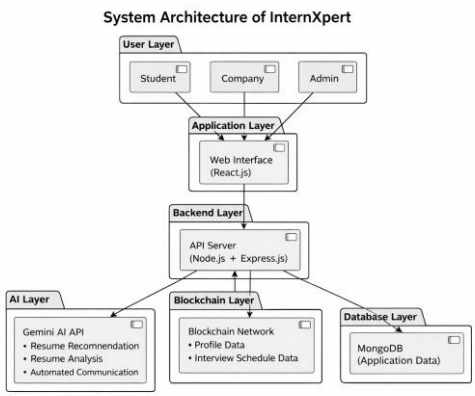


Figure 1 System Architecture of InternXpert

The development of the Intern-Xpert system follows a structured methodology integrating AI-driven intelligence, blockchain-based security, and AI-assisted communication.

5.1 Requirement Analysis

This phase involves collecting requirements for students, companies, and administrators. Key objectives include secure profile management, intelligent internship recommendations, real-time application tracking, and automated communication. Technologies are selected as

follows: React.js for the front-end, Node.js/Express.js for the back-end, MongoDB for operational data, Blockchain for storing sensitive information such as profiles and interview schedules, and the Gemini AI API for resume analysis, internship matching, and AI-powered chat.

5.2 Database Design

The database is designed to efficiently manage users, applications, internships, and company data. Critical information—such as student profiles and interview schedules—is stored on the blockchain, ensuring immutability, tamper resistance, and auditability. Non-sensitive data, including application status, notifications, and logs, are stored in MongoDB for fast querying and scalability.

5.3 Development of Backend Services

The backend is developed using Node.js with Express.js. It handles authentication, API endpoints, business logic, and secure integration with Blockchain. The Gemini AI API is incorporated for resume parsing, skill extraction, intelligent internship recommendations, and AI-assisted chat to handle student–company communications. Every critical operation on profiles and interview schedules is recorded on the blockchain to guarantee transparency and data integrity.

5.4 Frontend Development (React.js)

The front-end dashboard allows students to manage profiles, apply for internships, and receive AI-driven recommendations. Companies can post opportunities, review applications, and schedule interviews. Real-time updates, notifications, and AI-powered chat are integrated to ensure smooth communication between students and recruiters.

5.5 Integration and Real-time Features

All modules—including blockchain storage, Gemini AI services, and AI chat—are integrated to work seamlessly. Automated notifications and AI chat updates inform students about application status, interview schedules, and results. The blockchain layer guarantees that sensitive data remains tamper-proof, while AI provides intelligent guidance and real-time assistance.

6 MODULES AND ITS IMPLEMENTATION

The Intern-Xpert system consists of three main modules: Admin**,** Student**,** and Company, along with supporting modules such as Home, Registration/Login, and AI-powered Chat. Each module ensures security, transparency, and intelligent assistance.

6.1 Home Module: The landing page provides an overview of the platform and navigation to other modules. It includes summaries of available internships, company information, and highlights AI-powered features such as **intelligent recommendations** and **AI chat assistance.**

6.2 Register Module: Allows new users (students or companies) to create accounts by providing essential details such as name, email, and password. Admin verification ensures authenticity, and sensitive information like profiles and company verification data is stored on the **Blockchain** to prevent tampering.

6.3 Login Module: Enables returning users to securely log in using their credentials. Authenticated users access personalized dashboards, view applications, or post internships depending on their role.

6.4 Admin Module:

- **Login:** Admins authenticate using email and password.
- **View Students / Companies:** Admins can see all registered students and companies, including details and active internships.
- **Verify Students and Companies:** Admins verify authenticity, with verified records stored on the **Blockchain**.
- **Monitoring:** Admins monitor platform activity, including applications, postings, and system usage.
- **Logout:** Securely ends the admin session.

6.5 Student Module:

- **Registration / Login:** Students register with personal details and credentials or log in securely.
- **Profile Management:** Students update information and upload resumes. Profiles are stored on the **Blockchain** for security, and resumes are analyzed using the **Gemini AI API** to highlight skills and competencies.
- **View / Search Internships:** Students browse internships, with intelligent recommendations
- **Apply Internship:** Students submit applications; critical application data is recorded on the **Blockchain**, while operational details are in **MongoDB**.
- **Application Status Tracking:** Students track applications.
- **Attend Interviews:** Students view and confirm interview schedules, stored securely on the **Blockchain**.
- **Check Updates:** AI chat provides real-time updates regarding interview results or announcements.
- **Logout:** Securely ends the student session.

6.6 Company Module:

- **Registration / Login:** Companies create accounts or log in securely; registrations are verified by Admins and stored on the **Blockchain**.
- **Add Internships:** Companies post internship opportunities, including role description, eligibility, and duration.
- **View Applications:** Companies review student applications.
- **Interview Scheduler:** Companies schedule interviews; interview data is stored on the **Blockchain** for immutability.
- **Send Mail / Notifications:** AI chat assists automated communication with students regarding interview schedules and selection results.
- **Update Status:** Application status (selected, rejected, pending) is updated in real-time for transparency.
- **Logout:** Securely ends the company session.

6.7 Chatbot Module: The AI-powered chatbot enables real-time interaction with users. It leverages the **Gemini AI API** to process queries, provide intelligent guidance on internships, application procedures, and system navigation, and assist with automated notifications.

6.8 Logout Module: Allows all users—students, companies, and admins—to securely log out, clearing session data and maintaining privacy.

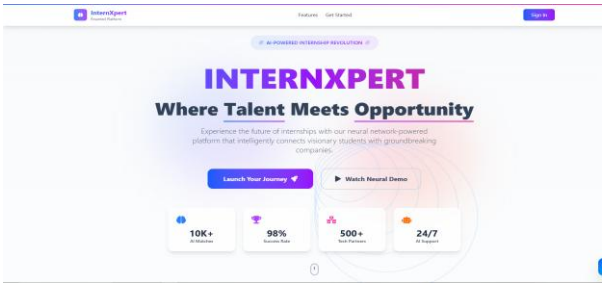
7 RESULTS AND DISCUSSION

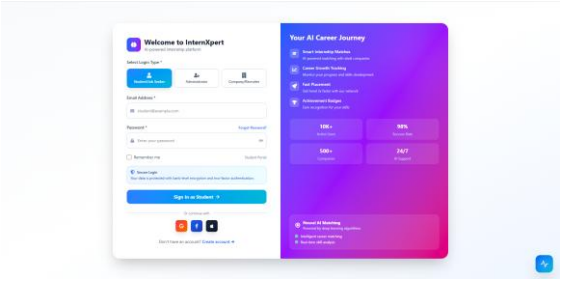
The Intern-Xpert platform integrates an AI-driven recruitment framework inspired by the ability-aware person–job matching approach proposed by Qin et al. in ACM Transactions on Information Systems (2020). Their model emphasizes understanding candidate abilities and job requirements using neural attention mechanisms to enhance matching accuracy. Building upon this concept, Intern-Xpert applies intelligent resume–job alignment to improve internship recommendations and candidate shortlisting. The system's landing page introduces these AI-powered capabilities, highlighting the platform's focus on precise skill matching, improved recruitment efficiency, and better career guidance outcomes compared to traditional keyword-based systems.

The user interface is designed to provide a seamless and secure experience across registration, login, and personalized dashboards. The registration and login modules ensure smooth onboarding for students, recruiters, and administrators while maintaining data security. Once logged in, users access role-based dashboards that present tailored functionalities and insights. Administrators manage platform operations, verify companies, and oversee postings to maintain authenticity and compliance. Companies can manage internship listings, update recruitment statuses, and review applicant profiles through structured dashboards that support organized hiring workflows and real-time monitoring.

From the student perspective, Intern-Xpert enhances engagement and transparency throughout the recruitment lifecycle. Students can maintain comprehensive profiles, upload resumes, and update career information to improve AI-driven matching accuracy. The platform also provides real-time application status tracking, allowing students to monitor whether their applications are pending, shortlisted, or rejected. Additionally, automated notifications and update features keep students informed about interview results, application progress, and important announcements. Together,

these features demonstrate how the proposed system improves communication transparency, recruitment efficiency, and user trust while delivering measurable advancements over conventional internship management platforms.





8 CONCLUSIONS

In this paper, the Intern-Xpert system was proposed as a secure, intelligent, and efficient solution for managing internships. The integration of Blockchain technology ensures that sensitive data, including student profiles and interview schedules, is tamper-proof and verifiable, enhancing trust and transparency. The use of the Gemini AI API enables intelligent internship recommendations, automated resume analysis, and AI-assisted communication between students and companies, improving decision-making and user experience. The system's modular design—comprising Admin, Student, and Company modules—facilitates seamless interactions, real-time application tracking, and efficient workflow management. Automated notifications and AI chat reduce manual effort while maintaining accurate and timely communication. The implementation using React.js, Node.js, Express.js, and MongoDB provides a scalable and responsive web platform.

By combining AI-driven assistance with blockchain-based security, Intern-Xpert addresses key challenges in online internship platforms, such as data authenticity, fraudulent activities, and inefficient coordination. The platform enhances recruitment efficiency, ensures transparency, and builds stakeholder confidence. Future work may include integrating predictive analytics for candidate success, advanced AI-driven interview scheduling, and support for cross-institution collaboration. Overall, Intern-Xpert represents a significant advancement toward secure, intelligent, and automated internship management systems.

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